

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Neural Computation and Applications [IT_4404]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) How does the functioning of the human brain influence the design and working principles of artificial neural networks in detecting and interpreting patterns? (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) A voice recognition system learns to understand accents better as it listens to more people speaking. Over time, it improves accuracy without being reprogrammed. (5)
What are the different types of learning processes in neural networks? Explain each with a suitable example.
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- 2) Apply the Perceptron Model to classify input data and explain how its key components contribute to the decision-making process. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Explain the relationship between the Perceptron and Bayes Classifier for a Gaussian environment. (5)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) In a weather prediction system, past data is combined with current observations to make accurate forecasts. Discuss how MAP Estimation helps in improving such predictions. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) A signal processing unit receives mixed signals containing both desired and interfering components. Explain how the filtering structure is designed to separate useful signals from noise. (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Apply the Backpropagation Algorithm to explain how weights are adjusted in a Multilayer Perceptron (MLP) during the learning process. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Analyze how the Radial-Basis Function Network (RBFN) differs from a Multilayer Perceptron (MLP) in terms of structure and learning approach (5)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Demonstrate how word embedding techniques such as Word2Vec or GloVe represent words as vectors in a neural network model. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Examine the differences between LSTM and GRU architectures in handling long-term dependencies in sequential data. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4404:Neural Computation and Applications

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4404:Neural Computation and Applications

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	5	25(50%)
4	Analysis	3	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)